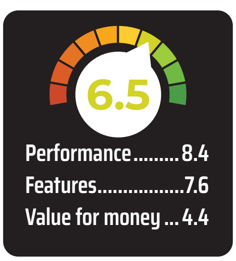
AMD Ryzen 9 9950X3D

An excellent processor with a big price tag

₹70,000

With the AMD Ryzen 9 9950X3D, Team Red has completed the launch of the high-end segments within the new 9000 series processor, albeit it does seem like that older non-X3D processors might be passed over considering how good the X3D processors have been performing. And with the competition lying low, the time is right for AMD to carve an even larger chunk of the desktop processor market share. We've reviewed the AMD Ryzen 7 9800X3D previously, and we were quite impressed by the way it managed all sorts of different workloads, from gaming to content creation. Although, there were a few workloads where either the competition shone or AMD's own higher core-count processors took the crown. With the AMD Ryzen 9 9950X3D, we expect AMD to knock it out of the park in practically every scenario. Let's find out if the 9950X3D is indeed the pinnacle of AMD's Zen 5 family.



SPECIFICATIONS

In a way, the AMD Ryzen 9 9950X3D is practically the same as the AMD Ryzen 9 7950X3D in most aspects. Obviously, the improved architecture does bring several enhancements, including the improved L1 cache, slightly better base clocks and support for higher memory speeds. Unfortunately, there's also a much higher TDP of 170 Watts for the 9950X3D which makes it a power-hungry processor. The lower base clocks compared to the 9800X3D will mean that the performance of the 9950X3D will be lower in quite a few benchmarks but the higher TDP might compensate for that and there's the possibility that you might get some extra performance. We'll let the benchmarks do the talking.

With all three Cache memory tiers considered, we do get a higher amount of Cache on the 9950X3D and that will be beneficial in some gaming work-loads. The lack of a dedicated NPU seems to have been the biggest loss in the sea of noteworthy improvements. Although, adding an NPU on a 16-Core processor would have taken the TDP up by quite a bit. Perhaps, we'll see that in the next generation.

SYNTHETIC PERFORMANCE

In Cinebench 2024, the AMD Ryzen 9 9950X3D scores quite high with a multi-threaded score of 2317 which takes it to the 2nd spot right behind the Intel Core Ultra 9 285K. And in the sing-threaded run, we see it in the same position with a score of 144. In Blender, the popular 3D rendering and animation software, the 9950X3D takes the top spot from Intel. The same behaviour can be seen across several content creation software. This essen-tially states that the Ryzen 9 9950X3D is an excellent processor for content creation workloads.

Moving to day-to-day productivity tasks, we see varying results with the 9950X3D but most often than not, it is always in the Top 3 processors in our benchmarks. In browser bench-marks, office applications, encryp-tion workloads, etc. the AMD Ryzen 9 9950X3D performs exceptionally well. There are some benchmarks in which the 9800X3D comes out ahead of the 9950X3D and we all know the reason why - chiplet design.

GAMING PERFORMANCE

Chiplet design has a lot of benefits